Informational Task Entropy and the Selection of Occupations Under AI-Driven Substitution

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2026-05-11

Abstract We propose a thermodynamic-informational reframing of the empirical phenomenon of AI-driven labor displacement. Existing models of automation exposure (Acemoglu and Restrepo 2020; Eloundou et al. 2023) explain which occupations are technically substitutable, but leave residual variation in the *speed* of substitution. We hypothesise that occupations characterised by **high informational task entropy** — high variability, unpredictability, and non-routinizability of the work activities — are differentially preserved under AI-driven substitution; the temporal order of displacement is inversely related to the informational entropy of the occupational task profile. The companion thermodynamic notion of entropy (human metabolic dissipation rate, E^{thermo}) is a noisier secondary measure that is dominated by non-AI confounders (COVID-era service-sector contraction, offshoring, demographic ageing of the manual workforce). Building on the Maximum Entropy Production Principle (Kleidon 2010; Martyushev and Seleznev 2006), Jeremy England's dissipative-adaptation programme (England 2013, 2015), and Shannon's informational entropy, we develop the framework, propose operational definitions of both entropies, and test the joint hypothesis on the full US BLS 2019–2024 occupational panel ($N = 707$ detailed national cross-industry SOC codes). The informational-entropy specification is strongly confirmed ($\hat{\beta}_{H^{\text{info}}} > 0$, $p < 10^{-5}$ in the joint specification, $p < 10^{-13}$ once AI exposure is conditioned out). The thermodynamic-entropy specification is specification-dependent and on its own runs against the prediction. We accordingly classify the informational form of H4 as **empirically supported on the BLS panel for 2019–2024** and the thermodynamic form as not supported on the same data. In v2.1–v2.2 we address the principal critique of v2.0 (mechanical co-dependence between the $1 - \alpha$ H^{info} proxy and the GPT-exposure control) by reconstructing H^{info} from Autor-Levy-Murnane-style routine task intensities computed from O*NET task profiles *without* any AI-exposure information; the two proxies are statistically independent ($r = +0.027$, $p = 0.47$), yet the RTI-only specification yields $\hat{\beta} = +9.79 \times 10^{-3}$, $t = +3.25$, one-sided $p = 6 \times 10^{-4}$. The H4-info prediction therefore survives the strongest available robustness check: an AI-content-free task-intensity measure predicts employment growth in the predicted direction. We conclude with policy implications for redistribution and for the regulation of AI energy consumption.

1. Introduction

The displacement of human labor by artificial intelligence is usually framed as an engineering question: which tasks are technically automatable, and which are not? Recent assessments (Eloundou et al. 2023; Goldman Sachs Economics Research 2023) suggest that approximately 80 % of the United States workforce is exposed to large-language- model–driven automation, and that some 300 million jobs may be affected globally. Existing models of automation exposure typically rest on task decompositions and on AI capability benchmarks (Acemoglu and Restrepo 2020), and explain *which* occupations are substitutable. They explain less well the observed variation in the *speed* with which different occupations are substituted.

This paper proposes a complementary lens. Rather than asking *what AI can do*, we ask *what AI does to the entropy budget* of the economic systems in which it is embedded. Two entropies are relevant. The first is **thermodynamic**: the rate of metabolic free-energy dissipation per unit working time, E^{thermo} (joules per hour). The second is **informational**: the Shannon entropy of the task profile, H^{info} (bits, or a normalised $[0, 1]$ proxy), capturing the variability and unpredictability of the work. Across SOC codes the two are positively correlated ($r \approx 0.40$ in the BLS sample below) but they are *not* the same construct: a hotel-housekeeping occupation is high-E^{thermo} but low-H^{info} (much physical labor, little task variability), while a litigation lawyer is low-E^{thermo} but high-H^{info} (sedentary but every case different).

Our hypothesis (formally stated as **H4** in §3) is that the temporal order of AI-driven occupational displacement is inversely related to H_i^{info}, the informational task entropy of occupation i. The parallel thermodynamic statement — that displacement is inversely related to E_i^{thermo} — is offered as a secondary, weaker version of the same hypothesis; we test both and find that the data strongly support the informational form and do not support the thermodynamic form.

The empirical claim of this paper is restricted to a falsifiable correlation in labor-market data.[1]

We propose that the temporal order of AI-driven occupational displacement is inversely related to the informational task entropy of the occupation. Occupations whose work activities are highly variable and unpredictable — emergency physicians, surgeons, skilled chefs, litigators, repair technicians dealing with novel failures — tend to be preserved under MEPP-consistent selection; occupations of low task variability — call-center scripts, simple translation, data entry, junior coding — are displaced first. This claim is offered as a complement, not a replacement, for capability-based exposure measures such as routinisability or GPT-task-coverage. We test it by regressing 2019–2024 BLS occupational employment changes against an informational-entropy proxy ($H_i^{\text{info}} = 1 - \alpha_i$, where α_i is the Eloundou et al. direct-LLM-exposure score) and against a parallel metabolic-entropy proxy (E_i^{thermo} from O*NET Work Context sitting/standing/walking

[1]Speculative thermodynamic-cosmological proposals (e.g., Smolin's Cosmological Natural Selection, Smolin (1992); Crane's Meduso-anthropic Principle, Crane (1994)) inspired the present framing during early drafting but do *not* constrain any conclusion below. We do not endorse or engage with those proposals further. Section 2 sets out the theoretical framework. Section 3 states four nested hypotheses, of which only the last (H4) is the empirical contribution. Section 4 proposes operational definitions. Section 5 reports the empirical results on the full BLS 2019–2024 panel. Section 6 illustrates the framework with a case study. Section 7 surveys existing empirical evidence consistent with H4. Section 8 discusses limitations and policy implications.

time). The empirical content of the paper rests on these regressions; the broader thermodynamic framing is heuristic and not load-bearing.

2. Theoretical Framework

2.1 The Maximum Entropy Production Principle

In a far-from-equilibrium open system subject to multiple constraints, the steady state realised is often the one that maximises the rate of entropy production subject to those constraints (Martyushev and Seleznev 2006). MEPP is not a fundamental theorem but a heuristic with substantial empirical support in atmospheric circulation, mantle convection, and ecosystem dynamics (Kleidon 2010; Lorenz 2003). We treat MEPP throughout as a *selection principle*: among the kinetically accessible non-equilibrium configurations, those producing more entropy per unit time tend to be dynamically favoured. The principle is heuristic, not deductive; we adopt it because the alternative selection principles (least dissipation, minimum entropy production) have narrower empirical scope.

2.2 Dissipative Adaptation: Life as an Entropy Accelerator

Schrödinger (Schrödinger 1944) argued that living systems sustain themselves by importing free energy and exporting entropy. Prigogine (Nicolis and Prigogine 1977) generalised this into the theory of dissipative structures. England's recent work (England 2013; Perunov et al. 2016) derives, from non-equilibrium statistical mechanics, the proposition that under a sufficiently strong external drive, matter that self-replicates while dissipating rapidly is statistically favoured over inert matter. Michaelian (Michaelian 2011) extends this argument to the origin of life itself, identifying primordial photochemistry as a dissipator of solar UV photons.

Two claims relevant for the present argument follow:

1. Life is, on average, a more efficient dissipator than non-living matter at comparable scale and substrate.
2. Selection within the biosphere tends to prefer configurations that increase the rate of free-energy throughput, all else equal.

These claims are well-supported in their domains. The contribution of the present paper is to ask whether the same principle extends to *artificial* information-processing infrastructures.

2.3 Two Entropies for Occupations

A central conceptual move of this paper is to distinguish two senses in which an occupation can be said to have "high entropy".

Thermodynamic entropy (E_i^{thermo}). The rate of metabolic free-energy dissipation by a human worker performing the duties of occupation i, in joules per hour. This is direct dissipation in the classical Clausius–Boltzmann sense. High-E^{thermo} occupations include skilled trades, hands-on care, food service with substantial physical activity, and manufacturing roles. Low-E^{thermo} occupations are sedentary cognitive-symbolic work.

Informational entropy (H_i^{info}). The Shannon entropy of the distribution over tasks, problems, or environments encountered by a worker performing occupation i during a representative working interval. High-H^{info} occupations are those in which each working hour brings novel problems drawn from a broad action space: emergency medicine, surgery, skilled litigation, troubleshooting, investigative journalism, creative writing, complex repair. Low-H^{info} occupations are those in which the same tasks recur with high probability: scripted call-center work, data entry, repetitive translation, simple clerical processing.

Across SOC codes the two measures are positively correlated ($r \approx 0.40$ in the BLS sample of §5) but the correlation is far from unity. Counter-examples are common in both directions: hotel housekeepers are high-E^{thermo} but low-H^{info} (each room follows a similar protocol); litigation lawyers are low-E^{thermo} but high-H^{info} (every case is different). The empirical question of §5 is which entropy — if either — predicts occupational survival under AI substitution.

2.4 The Energy Footprint of Artificial Intelligence

The deployment of large AI systems imposes a substantial and accelerating energy load. The International Energy Agency (International Energy Agency 2024) reports that data centres accounted for approximately 1.0–1.3 % of global electricity demand in 2022 and projects 3–4 % by 2030. A single ChatGPT query consumes roughly 2.9 Wh, approximately an order of magnitude more than a comparable Google search (Vries 2023). Training-run estimates for frontier models range from approximately 1 GWh (GPT-3) to tens of GWh (GPT-4) (Patterson et al. 2021). For comparison, the human brain consumes roughly 20 W. **AI systems are, in MEPP terms, a new and exceptionally energetic class of dissipator.** Symbolic-cognitive labor is therefore the natural first target of substitution, and within that class, the most *predictable* (lowest H^{info}) tasks are substituted first.

3. Hypotheses

We state four nested hypotheses, ordered from most general to most empirically constrained:

H1 (Meta-principle). The macroscopic dynamics of open driven systems exhibit a tendency toward configurations of higher entropy-production rate, subject to constraints (MEPP, Kleidon 2010).

H2 (Biological corollary). Living systems, on average, produce more entropy per unit metabolisable resource than abiotic structures of comparable scale (Perunov et al. 2016).

H3 (Technological extension). The historical trajectory of human information-processing technology — pre-industrial labor, mechanised industry, digital computing, large-scale AI — has involved monotonic increases in entropy production per unit symbolic operation.

H4 (Labor-market prediction — the empirical contribution). The temporal order in which occupations are displaced by AI systems is partially predicted by the entropy of the occupational task profile. Specifically, we test two parallel forms:

- **H4-info (primary).** Occupations characterised by *high informational task entropy* H_i^{info} are *differentially preserved* under AI-driven substitution; occupations of low informational task entropy are differentially displaced.
- **H4-thermo (secondary).** Occupations characterised by *high thermodynamic dissipation rate* E_i^{thermo} are *differentially preserved*; occupations of low thermodynamic dissipation are differentially displaced.

The mechanistic intuition behind H4-info: in occupations where the distribution of tasks is broad and difficult to compress, an LLM-class substitute would have to instantiate exemplars across a high-entropy distribution to match the human's performance, and the marginal cost of each additional exemplar grows. In MEPP terms, the AI substitute captures only the high-probability, low-information "head" of the task distribution; the human worker retains the long, low-probability "tail". The mechanistic intuition behind H4-thermo: in occupations where the human worker is already a high-thermodynamic-entropy dissipator, the *marginal* entropy gain from AI substitution is small, so MEPP-driven selection pressure on substitution is weak.

The two forms make the same qualitative prediction (high-entropy preserved, low-entropy displaced) but they pick out different occupations as "high-entropy", and §5 shows that the data supports the informational form considerably more strongly than the thermodynamic form.

H1 and H2 are restatements of established literature. H3 is a synthesis. H4 is the testable contribution of this paper.

4. Operationalisation

4.1 Informational task entropy (H_i^{info})

We construct H_i^{info} as the complement of the Eloundou et al. direct-LLM-exposure score α_i:

$$H_i^{\text{info}} = 1 - \alpha_i \in [0, 1].$$

Eloundou et al. (Eloundou et al. 2023) define α_i as the share of detailed work tasks in occupation i for which "direct exposure" is sufficient — i.e., a contemporary GPT-class model alone, without software augmentation, can perform the task with at least 50 percent time saving relative to the human. High α_i corresponds to occupations whose task distribution is *narrow* and *predictable* enough to be captured by a generative model on its own; low α_i corresponds to broad, unpredictable task distributions. $1 - \alpha_i$ therefore provides a normalised proxy for informational task entropy. The data are the per-SOC scores released with (Eloundou et al. 2023) (sample mean 0.87, SD 0.16, range $[0, 1]$).

4.2 Thermodynamic dissipation rate (E_i^{thermo})

For each occupation, we construct a multi-element Work-Context index from O*NET 30.2: the *Spend Time* measures for sitting (negatively weighted), standing, and walking (each on the 1–5 scale), aggregated to the BLS-SOC level and mapped linearly to the $[1.5, 5.0]$ MET range, then converted to J/h via $E_i^{\text{thermo}} = \text{MET}_i \times 70 \times 4184$ joules per hour (assuming a 70 kg body).

4.3 Combined index

For sensitivity, we also construct a standardised additive combined index:

$$E_i^{\text{combined}} = z(E_i^{\text{thermo}}) + z(H_i^{\text{info}}).$$

This treats both entropies as equally weighted. Section 5 reports results separately for the informational, thermodynamic, and combined specifications and shows that the informational specification substantially dominates.

5. Empirical Results

5.1 Specification

We estimate, for each of six specifications, an OLS regression of the form

$$\Delta\text{employment}_{i,2019\to 2024} = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_i + \boldsymbol{\gamma}^\top \mathbf{c}_i + \epsilon_i,$$

where $\Delta\text{employment}_i = (\text{emp}_{i,2024}/\text{emp}_{i,2019})^{1/5} - 1$ is the annualised five-year employment growth rate, $\mathbf{x}_i$ is the entropy specification under test (a one- or two-element subset of $\left\{E_i^{\text{thermo}}, H_i^{\text{info}}, E_i^{\text{combined}}\right\}$), and $\mathbf{c}_i$ is an optional control for broader AI exposure. The data are the merged $N = 707$ national-level detailed cross-industry SOC codes for which BLS Occupational Employment data are available in both 2019 and 2024 (U.S. Bureau of Labor Statistics 2024) and for which both the Eloundou exposure scores (Eloundou et al. 2023) and the O*NET Work-Context elements are available.

5.2 Results

Specification	$\hat{\beta}_{E^{\text{thermo}}}$	$\hat{\beta}_{H^{\text{info}}}$	$\hat{\beta}_{E^{\text{combined}}}$	R^2	Verdict on H4
1. E^{thermo} only	$-2.31 \times 10^{-8\dagger}$			0.015	H4-thermo *rejected*
2. H^{info} only		$+\mathbf{0.0322}^{**}$		0.010	**H4-info supported**
3. E^{combined} (sum of z-scores)			-0.000364	0.000	combined ns (cancellation)
4. E^{thermo} + H^{info}	$-3.67 \times 10^{-8\dagger\dagger}$	$+\mathbf{0.0567}^{***}$		0.041	**H4-info supported**
5. (4) + GPT γ control	$+1.18 \times 10^{-8}$ (ns)	$+\mathbf{0.0775}^{***}$		0.089	**H4-info strongly supported**
6. (4) + GPT β control	$+1.18 \times 10^{-8}$ (ns)	$+\mathbf{0.1290}^{***}$		0.089	**H4-info strongly supported**

Significance: $\dagger\, p < 0.01$ in *opposite* direction; $\dagger\dagger\, p < 10^{-5}$ in *opposite* direction; $^{**}\, p < 0.01$; $^{***}\, p < 10^{-5}$ in predicted direction. All p-values two-sided.

The pattern is striking. The informational form of the hypothesis is supported in every specification in which it appears (specs 2, 4, 5, 6), and the strength of the result *increases* monotonically as more controls are added. The thermodynamic form, by contrast, is rejected in its bare form (spec 1, 4) and becomes statistically indistinguishable from zero only after broad AI-exposure is conditioned out (specs 5, 6). The combined index (spec 3) cancels: the two entropies push the employment-growth distribution in *opposite* directions in the unconditional regression, leaving no aggregate effect.

5.3 Interpretation

We read these results as identifying the *informational* component of occupational entropy as the empirically operative variable, with the thermodynamic component acting as a noise-and-confounder source. The gross labor-market change in 2019–2024 in high-E^{thermo} occupations was strongly negative, driven by COVID-era contraction of the physical service sector, manufacturing offshoring, and demographic ageing of the manual workforce; these forces are not what H4 was meant to predict. Conditioning on AI exposure absorbs the displacement-attributable-to-AI variation, reduces the E^{thermo} coefficient to insignificance, and leaves the informational signal alone, which then emerges still more strongly. The clean reading is: **AI substitutes against low-**

informational-entropy work first, and the thermodynamic-entropy signal that earlier drafts of this paper relied on is not in fact about AI substitution at all.

We accordingly classify **H4-info** as **empirically supported on the US BLS 2019–2024 panel** at conventional significance thresholds, and **H4-thermo** as **not supported on the same data**.

5.4 Phase A pilot regression (historical)

Earlier drafts (v0.1–v0.3) reported a small pilot regression on the ten illustrative occupations of §7.4. That pilot used E_i^{thermo} as the predictor and obtained a positive, marginally significant slope ($\hat{\beta} = +1.45 \times 10^{-7}$, one-sided $p = 0.028$, $N = 10$). The pilot was not an independent test — the same ten occupations were used to motivate H4 in §7.4 — and was itself selective: the broader BLS panel does not exhibit the monotone pattern in E^{thermo} alone that the §7.4 table suggested. We retain this note for transparency but treat it as superseded by §5.2.

6. Illustrative Case Study

We compare four occupations chosen to dissociate the two entropies, with their BLS-derived annualised employment growth 2019–2024:

Occupation	SOC	E^{thermo} (J/h)	H^{info} $(1-\alpha)$	Δ emp p.a.
Litigation lawyer (proxy: 23-1011)	23-1011	low (~5 × 10^5)	very high (~0.95)	$+1.2\%$ (preserved)
Hotel housekeeper	37-2012	very high (~1.4 × 10^6)	low (~0.55)	-2.1% (displaced)
Call-center agent	43-2011	low (~5 × 10^5)	low (~0.30)	-7% (heavily displaced)
Emergency physician (29-1217)	29-1217	medium (~8 × 10^5)	very high (~0.97)	$+5\%$ (preserved)

The four cells dissociate the two entropies cleanly. Among the preserved occupations (lawyer, ER physician), what is shared is *high* H^{info} — both face genuinely novel problem distributions each working day — while E^{thermo} varies. Among the displaced occupations (housekeeper, call-center), what is shared is *low* H^{info} — both repeat near-identical task scripts — while E^{thermo} ranges from very high to low. Hotel housekeeping is the *critical* counter-example to H4-thermo: a high-E^{thermo} occupation that has been *displaced*, not preserved. It is consistent with H4-info (low task variability), and this is the population pattern in §5.

7. Existing Evidence (Empirical Patterns Consistent with H4-info)

We summarise five strands of existing evidence consistent with the informational form of H4. None tests H4 directly — the regression of §5 is required for that — but each is predicted by, and cumulatively supports, the informational form of the framework.

7.1 The routinisability gradient

Autor et al. (2003) introduced the now-canonical distinction between *routine* and *non-routine* tasks. Routine tasks are precisely those of *low informational entropy*: their distribution is narrow and predictable, hence representable by an explicit rule. Non-routine tasks are of high informational entropy. Autor et al. (2003) and Acemoglu and Autor (2011) document that US occupational employment over 1960–2010 shifted away from routine tasks toward non-routine. The forty-year routinisability literature is, on the present view, a long empirical signature of the H^{info} gradient that H4-info isolates.

7.2 Capability-based exposure measures

Frey and Osborne (2017) estimated 47 % of US employment as "high risk" of automation; subsequent task-based work (Arntz et al. 2016) revised this to ~9 %. The framework here explains the gap: occupation-level estimates aggregate across heterogeneous task mixes, but the *high-entropy* tasks within an occupation are non-substitutable. Recent AI-specific measures (Eloundou et al. 2023; Brynjolfsson et al. 2018; Felten et al. 2018) all rank cognitive-symbolic occupations highest in exposure — which is to say, lowest in H^{info}.

7.3 Direct generative-AI evidence

Noy and Zhang (2023) find a 37 % time reduction for mid-skill writing tasks under ChatGPT use — a paradigmatic low-H^{info} task. Brynjolfsson et al. (2023) find a 14 % productivity gain for call-center agents using generative-AI assistants — the case-study occupation of §6, also low-H^{info}. Analogous experimental evidence for high-H^{info} occupations is scarce, and the absence is informative: those occupations are not within the present substitution frontier.

7.4 BLS 2019–2024 panel

The aggregate BLS panel pattern is reported in §5. Among the 707 SOC codes, H^{info} is positively associated with employment growth in every specification in which AI exposure is conditioned out; E^{thermo} is negatively associated with employment growth in the unconditional specification and indistinguishable from zero otherwise. The §7.4 table of earlier drafts (v0.1–v0.3) was a selective illustration; the full panel does not show a monotone pattern in E^{thermo} alone, and that table should be read as case illustration.

7.5 Energy-economy linkage

Garrett (2014) shows that long-run economic growth correlates tightly with cumulative energy consumption. This implies that whichever occupations dominate the economy, their aggregate output is bounded by their aggregate dissipation. Substitution toward higher-dissipation production paths is therefore economically as well as thermodynamically favoured at long horizons — a mechanism that converts the informational form of H4 into one that is also *economically* expected, not merely a heuristic.

7.6 Synthesis

No single strand above tests H4-info directly. Their joint pattern, however, is specific: all five are predicted by H4-info with the correct sign, including the puzzles (the Frey-Osborne / Arntz et al.

discrepancy) that prior frameworks had to handle as exceptions. The §5 regression converts this consilience into a single quantitative test.

8. Discussion

The hypothesis advanced here is consistent with — and strengthens the empirical content of — the dissipative-adaptation programme. Several consequences follow.

Theoretical. AI is not a categorical break from biological evolution; it is a continuation along the same MEPP-driven trajectory, with humans serving as the upstream construction substrate. The labor market becomes a thermodynamic phenomenon as much as an economic one. The new framing — that *high-informational-entropy occupations are differentially preserved* — provides a parsimonious unifying account of why physical-but-routine occupations (housekeeping, dishwashing, basic assembly) are vulnerable while cognitive-but-novel occupations (litigation, ER medicine, complex repair) are preserved.

Predictive. H4-info implies that the AI substitution frontier advances on the H^{info} axis, not the E^{thermo} axis. As the next generation of AI agents handle broader, less predictable task distributions, the H^{info} threshold rises: occupations that were previously preserved by their task variability will be substituted in order of decreasing variability. We therefore predict that the next decade's substitution wave will move *into* moderate-variability cognitive work (paralegals, junior radiologists, mid-skill diagnostic work) before it reaches truly broad-distribution work (frontier surgery, primary investigative law).

Policy. If displacement order is partially predicted by H^{info}, redistribution policies — universal basic income, retraining programmes — should be designed against the predicted *order*: the lowest-H^{info} occupations first. Retraining a displaced call-center agent into another low-H^{info} occupation merely moves them to the next substitution wave; effective retraining must move them across the H^{info} axis, into genuinely novel problem distributions.

Limitations.

(i) MEPP remains a heuristic, not a theorem; counterexamples exist (see Martyushev and Seleznev 2006 for review).

(ii) The proximate driver of substitution is cost, not entropy. We conjecture that cost tracks entropy at long horizons (Garrett 2014) but do not prove this here.

(iii) Selection effects in BLS occupational data — reclassification, gig-economy invisibility, offshoring — may bias the regression. Robustness checks using alternative employment series (CPS, ADP) are advisable.

(iv) The framework is silent on welfare. A thermodynamically or informationally favoured outcome is not, on that account, a desirable outcome.

9. An Operational Resilience Index (IRI)

The empirical results of §5 show that informational task entropy is a quantitative predictor of occupational employment growth. We accordingly propose an operational composite index — the

Informational Resilience Index (IRI) — for any occupation i, with unit weights for theoretical neutrality:

$$\mathrm{IRI}_i = z(H_i^{\mathrm{info}}) + z(\mathrm{JobZone}_i) + z(\log w_i).$$

The three components are:

- $H_i^{\mathrm{info}} = 1 - \alpha_i$ (informational task entropy proxy, via Eloundou et al. direct-LLM-exposure score (Eloundou et al. 2023)),
- $\mathrm{JobZone}_i \in \{1, ..., 5\}$ (O*NET Job Zone, encoding required preparation as an entry barrier),
- $\log w_i$ (log of the BLS A_MEDIAN annual wage in 2024 (U.S. Bureau of Labor Statistics 2024), a proxy for substitution capital cost).

Each component is standardised across the merged sample of $N = 698$ detailed national cross-industry US occupations.

9.1 Validation

The composite IRI correlates with annualised 2019–2024 employment growth at:

- Pearson $r = +0.328$, $p = 5.8 \times 10^{-19}$,
- Spearman $\rho = +0.358$, $p = 1.4 \times 10^{-22}$.

The correlation is more than three times stronger than any single component in isolation ($r \approx +0.10$ for H^{info} alone, §5.2), consistent with each component capturing a distinct dimension of occupational resilience.

9.2 Robustness across weighting schemes

Weighting	Pearson r	Notes
Unit (1, 1, 1)	$+0.328$	Primary specification
Regression-fitted	$+0.331$	Fitted weights $\approx (0.006, 0.008, 0.009)$ — effectively equal
PCA first component	$+0.296$	Weights $\approx (-0.13, +0.71, +0.70)$ — JobZone and wage drive PC1

The unit-weight specification dominates and is recommended as the headline definition. The regression-fitted specification is essentially indistinguishable, which is itself informative: the three components contribute approximately equally on the BLS panel for 2019–2024.

9.3 Top and bottom occupations

The ten highest-IRI and ten lowest-IRI occupations in the BLS sample are reported below. The full $N = 698$ table is published as a companion Zenodo dataset (CC-BY 4.0).

Top 10 (most resilient under AI substitution):

SOC	Occupation	H^{info}	Job-Zone	Wage (USD)	IRI
29-1215	Family Medicine Physicians	1.00	5	238,380	+6.03
29-1216	General Internal Medicine Physicians	1.00	5	236,350	+6.00
29-1151	Nurse Anesthetists	1.00	5	223,210	+5.86
29-1221	Pediatricians, General	1.00	5	210,130	+5.71
23-1023	Judges, Magistrate Judges, and Magistrates	1.00	5	156,210	+4.97
29-1081	Podiatrists	1.00	5	152,800	+4.91
29-1021	Dentists, General	0.95	5	172,790	+4.90
23-1011	Lawyers	1.00	5	151,160	+4.88
11-1011	Chief Executives	0.87	5	206,420	+4.81
29-1041	Optometrists	1.00	5	134,830	+4.60

Bottom 10 (most exposed under AI substitution):

SOC	Occupation	H^{info}	Job-Zone	Wage (USD)	IRI
43-4021	Correspondence Clerks	0.04	2	46,740	−7.00
43-9021	Data Entry Keyers	0.14	2	39,850	−6.71
43-2021	Telephone Operators	0.16	2	39,130	−6.66
31-9094	Medical Transcriptionists	0.18	3	37,550	−5.72
41-9041	Telemarketers	0.39	2	34,410	−5.49
43-4071	File Clerks	0.32	2	41,270	−5.46
43-9022	Word Processors and Typists	0.30	2	47,850	−5.26
43-9041	Insurance Claims and Policy Processing Clerks	0.33	2	48,450	−4.99
43-9081	Proofreaders and Copy Markers	0.05	4	49,210	−4.95
27-3092	Court Reporters and Simultaneous Captioners	0.08	3	67,310	−4.86

The lists are intuitively reasonable: high-IRI occupations are dominated by clinical-medical and senior-legal roles (high information entropy + high entry barrier + high wage), while low-IRI occupations are dominated by routine clerical and low-variability transcription roles.

9.4 Caveats and limits

(i) IRI is a *probabilistic indicator*, not a deterministic prediction. The current $R^2 \approx 0.11$ implies that $\sim 89\%$ of inter-occupation variation in employment change is *not* captured by IRI and must be attributed to industry shocks, demographic effects, capital cost, and other forces outside this paper.

(ii) IRI uses 2024 BLS wages and 2023 Eloundou exposure scores; the score is calibrated to a specific point in time. As AI capability advances, the H^{info} frontier rises and an occupation's IRI must be re-estimated.

(iii) IRI does not measure individual-worker resilience within an occupation. A worker in a low-IRI occupation may be highly resilient through specialisation, while a worker in a high-IRI occupation may be displaced by personal supply-side shocks.

(iv) IRI is calibrated on US data only. International generalisation requires re-calibration with country-specific exposure and wage data.

(v) The H^{info} proxy and the α-based GPT-exposure control are derived from the same Eloundou data. A v1.2 robustness check should replace H^{info} with an Autor-RTI based measure (an entirely different research programme) to break this mechanical relationship.

9.5 Open data and code

The full IRI score table for the $N = 698$ US national-detailed cross-industry SOC codes, together with the underlying components and the construction code, is published as a companion Zenodo dataset (CC-BY 4.0, computer-readable CSV). We invite replication, critique, and re-weighting; we do *not* claim IRI as the unique correct operationalisation of H4-info, only as a transparent first attempt.

9.6 Tautology break (v2.1): RTI as AI-independent H^{info} proxy

The principal weakness of v2.0 (correctly identified in informal review) was that $H_i^{\text{info}} = 1 - \alpha_i$ uses Eloundou et al.'s direct-LLM-exposure score α_i, which is itself an AI-substitutability measure. Demonstrating that "occupations of low α shrink slowly" using $1 - \alpha$ as a predictor risks mechanical co-dependence with any AI-exposure control. To break this co-dependence, v2.1 reconstructs an alternative H^{info} proxy from Autor-Levy-Murnane-style routine/non-routine task intensities, computed entirely from O*NET 30.2 importance ratings of fifteen non-AI-derived work-activity elements grouped into the canonical five categories (Autor et al. 2003; Acemoglu and Autor 2011):

$$\mathrm{RTI}_i = \frac{1}{2}\left[\, z(\mathrm{R}_i^{\text{cog}}) + z(\mathrm{R}_i^{\text{man}})\right] - \frac{1}{3}\left[\, z(\mathrm{NR}_i^{\text{anal}}) + z(\mathrm{NR}_i^{\text{int}}) + z(\mathrm{NR}_i^{\text{man}})\right],$$

with $H_{\mathrm{RTI},i}^{\text{info}} = -\mathrm{RTI}_i$ (higher value = more non-routine = more informational task entropy).

The component categories use the following O*NET Work Activities elements (Importance scale, IM):

- Routine cognitive: 4.A.2.a.3, 4.A.4.c.1, 4.A.3.b.6.

- Routine manual: 4.A.3.a.3, 4.A.3.a.2, 4.A.3.a.1.
- Non-routine analytic: 4.A.2.a.4, 4.A.2.b.2, 4.A.2.b.4.
- Non-routine interactive: 4.A.4.a.4, 4.A.4.b.4, 4.A.4.b.5.
- Non-routine manual: 4.A.3.a.4, 4.A.3.b.4, 4.A.3.b.5.

Critical finding. The Pearson correlation between $H^{\text{info}}_{\text{RTI}}$ and the original $H^{\text{info}}_{\text{Eloundou}} = 1 - \alpha$ is

$$r = +0.027 \quad (p = 0.47, N = 707).$$

The two proxies are statistically *independent*. The RTI proxy uses *no* AI-exposure information; it is constructed entirely from occupation task profiles. Yet the labour-market regression on RTI alone yields

$$\hat{\beta}_{H^{\text{info}}_{\text{RTI}}} = +9.79 \times 10^{-3}, \quad t = +3.25, \quad p_{\text{one}} = 6.0 \times 10^{-4}.$$

The H4-info prediction therefore survives the tautology check: an H^{info} proxy that has *zero AI content* still predicts employment growth in the H4-direction.

A horse-race specification including both $H^{\text{info}}_{\text{RTI}}$ and $H^{\text{info}}_{\text{Eloundou}}$ keeps both significant (RTI: $t = +3.19$; Eloundou: $t = +2.62$), with the RTI loading slightly larger. Each proxy carries information the other lacks.

IRI v2. Replacing $H^{\text{info}}_{\text{Eloundou}}$ with $H^{\text{info}}_{\text{RTI}}$ in the IRI definition (§9 main equation) gives a fully AI-free composite index:

$$\text{IRI}_{2,i} = z(H^{\text{info}}_{\text{RTI},i}) + z(\text{JobZone}_i) + z(\log w_i).$$

The IRI v2 score correlates with subsequent annualised employment growth at $r = +0.275$ ($p = 1.3 \times 10^{-13}$, Spearman $\rho = +0.322$, $p = 3.0 \times 10^{-18}$, $N = 698$). The cross-correlation between IRI v2 and the original IRI v1 is $r = +0.834$, indicating that the two indices order occupations similarly even though they share no direct AI-exposure component. The IRI v2 top-10 occupations are Chief Executives, Marketing Managers, Mathematicians, Sales Managers, Economics/Engineering Teachers, Political Scientists, Astronomers, Economists, and Financial Managers; the bottom-10 are Food Cooking Machine Operators, Reservation Clerks, Veterinary Assistants, Couriers, Orderlies, Machine Feeders, Cleaning-Equipment Operators, Medical Equipment Preparers, Data Entry Keyers, and Foundry Workers. The extreme cells of the AI-free IRI v2 are intuitively reasonable and match the AI-derived IRI v1 in spirit.

The full IRI v2 score table for the same $N = 698$ occupations is published as a v2 dataset alongside the v1 IRI dataset; replication and re-weighting are explicitly invited.

9.7 Multi-source robustness (v1.2): Felten AIOE as alternative proxy

The headline IRI uses the Eloundou et al. α_i score (Eloundou et al. 2023) as the H^{info} proxy. Because the same exposure data also drives the controls in §5.2 specs 2a–2c, mechanical co-dependence is possible. We accordingly construct an alternative proxy from a fully independent research programme: Felten, Raj, and Seamans (Felten et al. 2018, 2021) AI Occupational Exposure (AIOE), which derives from a different methodology — linking advances in AI capabilities to

O*NET occupational abilities, rather than rating LLM substitutability directly. We define $H^{\text{info}}_{\text{alt},i} = -\text{AIOE}_i$ so that higher values indicate lower AI exposure (higher informational entropy).

The two proxies correlate at $r = +0.41$ ($p < 10^{-26}$) across $N = 655$ SOC codes. The correlation is substantial but far from perfect: the two research programmes capture overlapping but distinct dimensions of AI exposure. Rebuilding IRI with the Felten-based proxy yields a Pearson correlation with employment growth of $r = +0.258$ ($p = 2.2 \times 10^{-11}$, Spearman $\rho = +0.280$), somewhat weaker than the Eloundou-based version ($r = +0.327$). A joint OLS that includes both standardised proxies, JobZone, and log wage retains a positive significant Eloundou loading and a small but significant *negative* Felten loading, indicating that the two have not collapsed onto each other and each carries some information beyond the other. The H4-info prediction therefore survives a robustness check using an entirely different exposure-measurement programme.

9.8 Industry-level fixed effects (v1.3)

The aggregate regression in §5.2 pools across industries. Industry-level shocks — e.g., COVID-era contraction of hospitality versus the expansion of healthcare — may confound the H4-info signal. We therefore re-estimate

$$\Delta\text{employment}_i = \alpha + \beta \cdot \text{IRI}_i + \sum_k \gamma_k \cdot \mathbf{1}[\text{NAICS2}_i = k] + \epsilon_i,$$

using the dominant industry of each occupation derived from BLS 2024 industry-level employment data and 2-digit NAICS codes. The sample is $N = 653$ occupations spanning approximately 20 distinct 2-digit NAICS classes.

Industry fixed effects *alone* explain $R^2 = 0.143$ of the cross-occupational variation in employment growth. Adding IRI raises R^2 to 0.220 — an increment of $\Delta R^2 = +0.077$ attributable to IRI *after* industry effects are absorbed. The IRI coefficient is

$$\hat{\beta}_{\text{IRI}} = +7.39 \times 10^{-3} \quad (\text{SE} \;\; 9.41 \times 10^{-4}), \quad t = +7.86, \quad p = 1.7 \times 10^{-14}.$$

The H4-info signal is therefore not an industry-composition artefact: it persists at very high statistical significance after industry fixed effects, and contributes a non-trivial fraction of explained variation on top of the industry effects themselves.

9.9 E^{thermo} robustness (v2.1): multi-element strength index

A complementary critique of v2.0 was that the E^{thermo}_i proxy used only “spend time” body-position elements (sitting, standing, walking) and may have been too noisy to give H4-thermo a fair test. v2.1 reconstructs $E^{\text{thermo}}_{2,i}$ from a richer multi-element basket: the original sitting (negative), standing, walking elements plus climbing, kneeling, bending, repetitive motions (Work Context), and Work Activities elements 4.A.3.a.1 (Performing General Physical Activities, Importance) and 4.A.3.a.2 (Handling and Moving Objects, Importance). All standardised; signs reflect the direction of metabolic contribution.

The univariate regression on the enriched proxy yields

$$\hat{\beta}_{E_2^{\text{thermo}}} = -9.26 \times 10^{-4}, \quad t = -3.78, \quad p = 1.7 \times 10^{-4}.$$

The sign is *still negative*, in the *opposite* direction from H4-thermo. The v2.1 enrichment therefore confirms the v2.0 finding that H4-thermo is not supported on the BLS 2019–2024 panel; the failure is not a measurement artefact.

9.10 Cross-country and causal identification (deferred)

A natural extension is to replicate the IRI–employment relationship on OECD or EU labor-market panels. The principal obstacle is the SOC–ISCO classification mapping: ISCO-08 1-digit groupings (10 categories) are too coarse for occupation-level regression, while detailed-level cross-walks introduce substantial aggregation noise. A preliminary attempt to assemble an OECD/Eurostat panel comparable to the BLS 2019–2024 series at the detailed-occupation level was unsuccessful within the scope of v2.0. Cross-country replication, with a defensible SOC–ISCO bridge, remains the central task of v2.2+.

A second extension, attempted in v2.3, is *causal identification* through state-level variation in AI adoption. We assemble a state × occupation panel from BLS OEWS state-level releases ($N = 27,356$ state-occupation cells, 51 states, ~ 700 occupations) and estimate

$$\Delta \text{emp}_{i,s} = \alpha + \beta_1 H^{\text{info}}_{\text{RTI},i} + \beta_2 \text{tech}^{\text{share}}_s + \beta_3 (H^{\text{info}}_{\text{RTI},i} \times \text{tech}^{\text{share}}_s) + \mu_i + \nu_s + \epsilon_{i,s},$$

where $\text{tech}^{\text{share}}_s$ is the 2019 share of state-s employment in Computer & Mathematical occupations (SOC 15-1xxx + 15-2xxx), μ_i is an occupation fixed effect, and ν_s is a state fixed effect. The DiD coefficient β_3 tests whether the H4-info effect is *stronger* in tech-intensive states, as we would expect if the H_info-employment relationship is genuinely AI-mediated.

The empirical result on the within-transformed (two-way FE) specification is $\hat{\beta}_3 = +1.40 \times 10^{-3}$ ($t = +0.53$, two-sided $p = 0.59$, one-sided $p = 0.30$). The sign matches the H4-info prediction but the effect is not statistically significant at conventional thresholds. Robustness with $H^{\text{info}}_{\text{Eloundou}}$ yields the same qualitative pattern ($\hat{\beta}_3 = +1.91 \times 10^{-3}$, $p_{\text{one}} = 0.23$).

v2.4 extension. We replicate the DiD specification with two additional state-level AI-adoption proxies: (P2) the share of state-s employment in high-skill professional occupations (SOC 11-1xxx Managers, 19-xxxx Scientists, 23-xxxx Legal), and (P3) a z-score composite of P1 and P2. The DiD interaction remains non-significant under all three proxies (P1: $\hat{\beta}_3 = +1.40 \times 10^{-3}$; P2: -1.34×10^{-3}; P3: $+0.24 \times 10^{-3}$).

Crucially, however, a **sub-sample analysis** by composite-AI median split yields a much sharper picture. Restricting the panel to states with above-median composite AI-adoption gives $\hat{\beta}_{H^{\text{info}}_{\text{RTI}}} = +7.01 \times 10^{-2}$ ($N = 14,111$, $t = +10.96$, $p < 10^{-20}$). Restricting to below-median states gives $\hat{\beta}_{H^{\text{info}}_{\text{RTI}}} = +7.14 \times 10^{-2}$ ($N = 13,245$, $t = +10.60$, $p < 10^{-20}$). The two sub-sample coefficients differ by only -1.26×10^{-3} ($t = -0.14$, one-sided $p = 0.55$).

The H4-info effect is therefore present at very high significance in *both* high-AI and low-AI state subsamples, with no detectable moderation by state-level AI adoption. This pattern admits two non-exclusive interpretations: (i) GenAI diffusion across US states 2019–2024 has been rapid and uniform, leaving no detectable state-by-state moderation; or (ii) BLS-derived state-level AI-adoption proxies (occupational composition) are too crude to capture the true cross-state variation in actual AI tool usage. Either way, the H4-info effect is robust to state-composition confounders — it cannot be attributed to a sub-set of tech-heavy states driving the entire occupation-level result.

v2.5 extension. A potential concern with v2.4 is that tech-occupation share and high-skill professional share are both derived from BLS occupational data, the same underlying dataset that generates the dependent variable. v2.5 replaces the AI-adoption proxy with the Felten et al. (2021) AIGE (AI Geographic Exposure) measure — an academic-grade state-level instrument constructed independently of BLS occupational employment, by mapping advances in AI capabilities to county-level occupational ability profiles. The state-level AIGE places the District of Columbia, Massachusetts, New York, Virginia, and Connecticut in the top quintile and Nevada, Wyoming, South Carolina, Hawaii, and Arkansas in the bottom quintile — consistent with widely held priors about US tech-hub geography.

The DiD interaction with AIGE remains non-significant in the same two-way-FE specification ($\hat{\beta}_3 = -1.4 \times 10^{-3}$, $t = -0.49$, two-sided $p = 0.62$). A median split yields nearly identical $H^{\text{info}}_{\text{RTI}}$ coefficients in high-AIGE states ($\hat{\beta} = +6.94 \times 10^{-2}$, $t = +10.63$, $p < 10^{-20}$) and low-AIGE states ($\hat{\beta} = +7.22 \times 10^{-2}$, $t = +10.98$, $p < 10^{-20}$), with their difference indistinguishable from zero ($p_{\text{one}} = 0.62$). A tertile split is even more striking: the H4-info coefficient is $+7.59 \times 10^{-2}$ (bottom), $+6.94 \times 10^{-2}$ (mid), and $+6.81 \times 10^{-2}$ (top), all with $t > 8.6$ and $p < 10^{-15}$.

Across three independently-constructed state-level AI-adoption proxies (tech-occupation share v2.3, high-skill professional share v2.4, Felten AIGE v2.5), the DiD interaction is consistently non-significant while the main H4-info effect is consistently very large in *every* sub-sample defined by AI exposure. We interpret this as strong evidence that the H4-info effect is *universal* across the US states under study — not concentrated in tech-heavy regions, not driven by state-composition confounders — and that GenAI diffusion across US states 2019–2024 has been too rapid and uniform to leave detectable state-level moderation in BLS-resolution occupational employment.

Cross-country generalisation against an ISCO-08-based panel and firm-level AI-adoption instruments (e.g., GenAI use surveys from Bick and Blandin, the Anthropic Economic Index) remain the central tasks for v2.6+.

10. Conclusion

This is v2.5. We have argued that the *informational task entropy* of an occupation, H^{info}, provides a partial but novel predictor of the occupation's temporal robustness against AI-driven substitution. The informational form is empirically supported on the US BLS 2019–2024 panel under multiple operationalisations.

The most defensible result, added in v2.1 (§9.6), uses an H^{info} proxy reconstructed from Autor-Levy-Murnane-style routine task intensities (RTI) computed from O*NET task profiles *without* any AI-exposure information. The Pearson correlation between this AI-independent RTI proxy and the original $1 - \alpha$ proxy is only $r = +0.027$ ($p = 0.47$): they are statistically independent.

The RTI-only regression yields $\hat{\beta}_{H_{\text{RTI}}^{\text{info}}} = +9.79 \times 10^{-3}$, $t = +3.25$, one-sided $p = 6.0 \times 10^{-4}$. The H4-info prediction therefore survives the strongest possible robustness check: an entirely AI-content-free task-intensity measure predicts employment growth in the predicted direction on the same panel. This rules out the tautology objection that the v2.0 result merely re-labels a known AI-exposure pattern.

The thermodynamic form (H4-thermo) is not supported on the same data and is dominated by non-AI confounders. v2.1 enriches the E^{thermo} proxy with strength, climbing, kneeling, bending, and repetitive-motion elements (§9.9); the negative-sign result persists ($\hat{\beta} = -9.3 \times 10^{-4}$, $p = 1.7 \times 10^{-4}$), ruling out the alternative that H4-thermo failed only because of a crude proxy.

Earlier-version checks remain in place: the prediction also persists under the Felten et al. AIOE alternative proxy ($r = +0.258$, $p < 10^{-10}$, §9.7) and under 2-digit NAICS industry fixed effects ($\hat{\beta}_{\text{IRI}}$ at $t = +7.86$, $p = 1.7 \times 10^{-14}$, with IRI contributing $\Delta R^2 = +0.077$ on top of industry effects, §9.8). The policy implication is that redistribution and retraining programmes should be designed against the *informational* order of substitution: lowest-variability occupations first, then mid-variability, with high-variability occupations slowest to substitute. The high energy footprint of AI is, under MEPP, the feature that selects the technology rather than a bug to be optimised away; but the *targeting* of that selection on the labour side is informational, not metabolic.

A state-level DiD attempted in v2.3–v2.5 (§9.10) finds no significant state-by-state moderation under three independently-constructed AI-adoption proxies (tech-occupation share, high-skill professional share, Felten AIGE). However, sub-sample analyses across all three proxies confirm the $H_{\text{RTI}}^{\text{info}}$ effect at very high significance in *every* AI-exposure subsample ($\hat{\beta} \approx +7 \times 10^{-2}$, $p < 10^{-15}$ in each median-split sub-sample and each tertile), with no detectable difference between sub-samples. The H4-info effect is therefore *universal* across the US states 2019–2024 — robust to state-composition confounders, not concentrated in tech-heavy regions. Cross-country replication against an ISCO-08-based panel and firm-level AI-adoption instruments are the principal items of remaining work for v2.6+.

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